

Ergonomic risk assessment of working postures of nurses working in a medical faculty hospital with REBA and RULA methods

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Abstract

The aim of the study is to reach objective results about the ergonomic risks of nurses' working postures not only by using questionnaires but also by using Rapid Entire Body Assessment (REBA) and Rapid Upper Limb Assessment (RULA) methods, which are risk assessment methods used in engineering fields. This study is a cross-sectional study. First of all, 383 nurses were evaluated with a survey. 91.4% of the nurses were women and 6.5% were men. Average age (female: 38.20; male: 31.5), average weight (female: 65.60 kg; male: 81.50 kg), average height (female: 162.7 cm; male: 176.80 cm) and average body mass index (female: 24.80; male: 26) were calculated. In the hospital, which we grouped into in the hospital, which we divided into eight workstations, the REBA and RULA scores of the three nurses working at each workstation were calculated and the average REBA and RULA scores of the workstations were determined. The percentage of those who experienced musculoskeletal pain in the last 1 year at each workstation and the mean REBA–RULA scores were as follows; ward (92%; 6.92–5.69), operating room (98%; 10–7), emergency department (100%; 6.08–5.33), outpatient clinic (75%; 5.5–5.16), intensive care (94%; 6.21–5.57), administrative affairs (94%; 5–5), laboratory (95%; 5.80–5.2) and interventional procedure (92%; 6.4–5.4). In these stations, the highest scores

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were seen in the operating room. In the questionnaire, 92.46% of the nurses reported having complaints in at least one part of the musculoskeletal system in the last year. The hospital weighted mean score was REBA 6.86 and RULA 5.71. When REBA and RULA measurements were evaluated, it was determined that the risk in the departments where the nurses worked was generally at the medium risk level. REBA and RULA scores indicate that ergonomic changes and adjustments should be made. For this determination, the risks of working postures should be determined with quantitative methods.

Keywords

Ergonomics, musculoskeletal system, REBA, RULA, nurse

Introduction

Studies conducted in recent years have shown that there has been an increase in work-related musculoskeletal diseases (WRMD), which are accepted as occupational diseases in many industrializing countries.

In a study involving 6500 people in Finland, 35.1% of people had chronic pain.¹ In an article published in 2016, it was stated that 50% of the work-related ailments constituted WRMD, and today it ranks third in terms of applying to a health institution due to these ailments.²

The incidence of WRMD, announced by the U.S. Bureau of Labor Statistics in 2020, was 25.4 per ten thousand.³ At least one WRMD was detected in 19.7% of those who applied to the occupational diseases outpatient clinic of a university hospital in Turkey between 2015 and 2018.⁴ According to the World Health Organization, an average of 1.71 billion people in the world have musculoskeletal disorders and they think that this number will increase.⁵ According to the data of the European Agency for Occupational Health and Safety, the loss of incapacity life years (DALY) due to the WRMD in 2019 constituted 15% of the total.⁶ Health sector working conditions are more stressful and problematic when compared to different business lines. It has been shown that the quality of working life of nurses who are health workers is also related to musculoskeletal diseases.^{7,8}

At the point of work-related musculoskeletal system diseases, more research is being carried out in the field of health, as in all business sectors, to reduce the frequency of musculoskeletal system diseases in the world and in Turkey, to detect the changes that have occurred over the years and to make improvements.⁹

Ergonomic risk analysis methods are technically divided into three categories. These are the recording method, systematic observational methods and direct measurement methods.¹⁰ Rapid Entire Body Assessment (REBA) and Rapid Upper Limb Assessment (RULA) are systematic observational methods. Measurement evaluation with video recordings falls under advanced observation methods. Finally, there are risk analyses that are called direct measurement methods, that is, measurements made with optical instruments attached to the human body.¹¹

With these new and advanced methods other than questionnaire scales, pictures are taken according to the posture of the people in their work, and parameters such as extremity movements, joint angles and the loads they are exposed to are determined by the standard scores and more realistic body risk analyses are revealed.¹²

REBA is preferred because it is suitable for workplace use, fast, has practical results, is economical, and does not require a device that can disturb the employee and prevent the execution of the work. This method is a simple observation-based method designed by McAtamney and Hignett in 1999 and is designed to be sensitive to the types of unpredictable working postures that exist in other service sectors, especially in the healthcare sector.^{13–15}

One hundred and forty-four posture combinations were coded independently by ergonomists and physiotherapists, and then load, grip, and activity scores were combined to produce the REBA outcome score, which determines risk and action levels.¹⁵

In 1993, McAtamney developed RULA to analyze the working postures of the trunk and legs, that is, neck, upper arms, lower arms and wrists. It is a method that evaluates the risks of different ergonomic impacts in terms of WRMD. It is a simple and fast observational method used in workplaces.^{16,17}

RULA is one of the two commonly used analysis methods together with REBA to study economic reasons and ease of use.¹⁸ RULA assessment is performed on the upper and lower parts of the body. It is mostly used to measure body risks for sedentary and laid jobs.¹⁹ The ideal working posture in terms of its effects on the employee is a working posture that pays attention to the rules according to the posture used.²⁰ Work posture is an aspect of worker safety.²¹ Reports and research conducted in domestic and foreign industrial sites draw attention to musculoskeletal disorders. It is known that the disorder not only causes economic problems but also affects production efficiency. Thanks to such studies, work effort will be reduced and worker safety and productivity will be increased.²²

The purpose of this study conducted on nurses is to include REBA and RULA, which are risk assessment methods based on systematic observation and mostly used in engineering fields, in addition to the survey application. Thus, original, more reliable and generalizable results and evaluations will be achieved in the research.

Methods

The study was carried out with nurses working at Istanbul Medical Faculty Hospital. Ethics committee approval was obtained from the Istanbul University Istanbul Faculty of Medicine Ethics Committee with the date of October 17, 2017 and decision number 1140.

The research materials and methods consist of three parts. The first of these will be explained about the nurse participant form. Demographic information and complaints about the musculoskeletal system were questioned in the questionnaire form. In our study, Erdinç et al. applied the Turkish validity and reliability of Cornell's "Turkish version of the Cornell Musculoskeletal Discomfort Questionnaire: Cross-cultural adaptation" applied to nurses had values of 0.89.²³

The research population consisted of 862 nurses working at Istanbul Medical Faculty between December 2017 and March 2018. In the literature review, it is understood that approximately 40–60% of healthcare professionals have musculoskeletal disorders.^{24–26}

Accordingly, the sample size was found to be 334 individuals with 50% incidence 95% confidence interval (type I error 0.05, power 0.80) and 6% deviation.²⁷

$$n = \frac{(N(Z_{\alpha} + Z_{\beta})^2 pq)}{(d^2(N - 1) + (Z_{\alpha} + Z_{\beta})^2 pq)}$$

$$n = \frac{(862 * (1.96 + 0.842)^2 * 0.50 * 0.50)}{((0.06)^2 * (862 - 1) + (1.96 + 0.842)^2 * 0.50 * 0.50)} = 334$$

where N is the number of individuals in the universe; n is the number of individuals to be sampled; p is the incidence of the event under investigation; q is 1-p; Z_{α} , Z_{β} are fixed values taken for alpha error and strength of test; d is the acceptable sampling error.

The forms were applied to 402 nurses instead of 334, taking the probability of the nurses working in the hospital to fill in the questionnaires incompletely or not answering them as 20%. When the questionnaire forms were examined, it was determined that there were 19 nurses who did not answer the basic questions, and these were not included in the evaluation. The study was carried out with the participation of 383 nurses.

Before the interview, the researcher introduced herself, stated her purpose and what she wanted to study with them, and informed the nurses about the content of the research. In addition to verbal information, the researcher obtained written consent by filling out the form by creating a section in the questionnaires stating that the necessary information about the research, approval of the Ethics Committee and institutional permissions were created. Working as a nurse at the Istanbul Faculty of Medicine at the time of the study and obtaining voluntary consent were deemed sufficient as inclusion criteria. The exclusion criteria were that the nurse was on leave during the research period or did not give consent to the research and did not answer questions about the musculoskeletal system. The second and third method used in the research, apart from the survey method, was REBA and RULA score calculations.

In REBA, one of these methods, the whole body is evaluated and scoring is done according to the angular values and posture of the relevant part of the body during the posture to be analyzed. The scoring process of the risk caused by the posture or movement analyzed in the REBA method is calculated by following various steps.

With this method, a score ranging from 1 to 15 is determined according to the loads that may occur in the neck, trunk, legs, arms and wrists during work. As this score value progresses from 1 to 15, it is determined that the necessity and urgency of taking precautions increases.¹⁵

It also indicates that the risk level increases. At the last stage, the control measure level of the range corresponding to the REBA score is determined as shown in Table 1.¹⁵

RULA analyzes the working postures of the wrist, lower arm, upper arm, neck, trunk and legs. In the method, the human body is divided into two different groups, A and B. Group A includes the lower arm, upper arm and wrist parts and group B includes the neck, torso and leg parts. The postures of body parts are scored according to their angular range of motion.¹⁷

With the RULA method, musculoskeletal system problems that may occur in the individual can be detected by taking into account the power needed for the work to be done.

Table 1. REBA risk rating.

Degree	REBA score	Risk level	Prevention
0	1	Negligible	Not necessary
1	2–3	Low	It may be necessary
2	4–7	Middle	Necessary
3	8–10	High	Necessary soon
4	11–15	Very high	Needed immediately

Hignett and McAtamney.¹⁵**Table 2.** RULA score action levels.

Score	Result
1–2	It is an acceptable result if it is not repeated for a long time.
3–4	The work done should be analyzed better, some changes can be made.
5–6	It is necessary to review the work done as soon as possible and make changes in the short term.
7	There is an urgent need to review and change the work done.

McAtamney and Nigel Corlett.¹⁷

The effects that may occur on the individual due to working conditions are evaluated with a scoring system. RULA analysis examines the individual's movements, working posture and the risk factor of the load on the musculoskeletal system. The final score obtained after completion of the analysis may vary between 1 and 7. As seen in Table 2, final scores are evaluated by dividing into four different action levels.¹⁷

In this study, REBA and RULA scores were calculated at eight designated workstations in order to measure the body risks resulting from nurses' working postures. These workstations are the departments where "Services, Operating Room, Emergency Service, Polyclinic, Intensive Care, Desk Works, Laboratory and Interventional Procedures" are performed. The 19 jobs most frequently performed by nurses were identified and shown in Figure 1.

In addition, since the research was conducted in a hospital with very intensive patient services, no pictures could be taken from all nurses. REBA and RULA score calculations were made by taking pictures from a total of 24 nurses, 3 from each workstation.

In Figure 2, the working postures of the nurses while performing various tasks at the workstations are photographed.

In the statistical analysis phase, first, descriptive statistics (mean, standard deviation, median, minimum, maximum values and percentage ratio values) were calculated. For each pain region, the risk of pain according to the workstations was examined by logistic regression analysis and the odds ratio (OR) values for the back region, which were significant, were given with 95% confidence intervals. Statistical significance was accepted as two-way and $p < 0.05$. Statistical Package for Social Sciences 21.00 program was used for statistical evaluation.

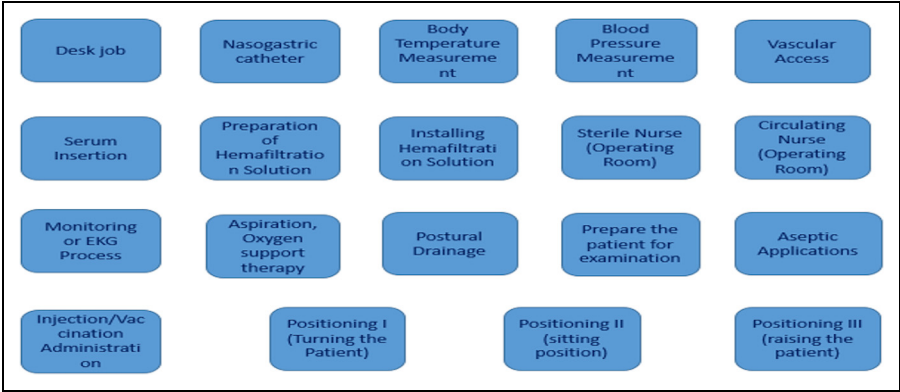


Figure 1. Work done at workstations.



Figure 2. Sample pictures where various works were done at various stations in our research.

Results

Of the 383 nurses participating in the study, 350 (91.4%) were female, 25 (6.5%) were male, and 8 (2.1%) nurses did not specify their gender. Similarly, descriptive statistics for weight, height and body mass index (BMI) variables were calculated over the respondents. Table 3 shows that the mean age of female nurses was 38.20 ± 9.09 years, and the average age of male nurses was 31.50 ± 8.80 years. Again, it was determined that the mean weight of the female nurses who said their weight measurements was 65.60 ± 11.39 kg, and the average weight of the male nurses who answered was $81.50 \pm$

Table 3. Age, weight, height and BMI distribution by gender:

Gender		Male (n:25)										
Women (n:350)												
Variable	n	Average	Sd.	Median	Max.	Min.	n	Average	Sd.	Median	Max.	Min.
Age (n:356)	331	38.20	9.09	38.00	64.0	22.00	25	31.50	8.80	28.0	64.00	23.00
Weight (n:364)	339	65.60	11.39	65.00	1100	43.00	25	81.50	12.40	80.0	103.00	61.00
Height (n:369)	344	162.70	5.92	163.00	1800	148.00	25	176.80	5.90	178.00	190.00	165.00
BMI (n:364)	339	24.80	4.29	24.00	42.4	17.40	25	26.00	3.7	25.60	34.80	19.90

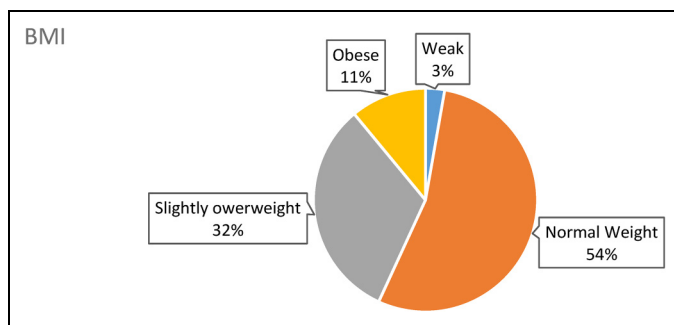


Figure 3. BMI graph.

12.40 kg. The average height of the female nurses who responded to the height measurements was 162.70 ± 5.92 cm and the average height of the male nurses was 176.80 ± 5.90 cm. In light of these data, the mean BMI was calculated to be 24.80 ± 4.29 in female nurses and 26 ± 3.7 in male nurses.

The BMI data of male and female nurses who shared their height and weight information are shown graphically in Figure 3. Three percent of the nurses were evaluated as underweight and 11% were evaluated as obese.

Table 4 shows the distribution of musculoskeletal complaints of nurses according to the workstations they work at. As seen in the table, neck pain was experienced in the emergency services with a maximum rate of 87.50%. Shoulder complaints were felt at the highest rate of 81.82% in interventional treatment procedures. Elbow complaints were mostly experienced in laboratories with a rate of 31.58%. Wrist complaints were observed with a rate of 63.64% at the workstation where interventional treatment was performed the most. It is seen in the table that back complaints are experienced at a high rate in almost all workstations, but the highest rate (100%) is experienced in interventional treatment procedures. Low back complaints were experienced at the highest rate of 94.44% in polyclinic nurses. Hip-thigh complaints were highest at the workstation where the interventional procedure was performed at a rate of 54.55%. Knee complaints were mostly experienced in emergency services with a rate of 62.50%. Foot complaints were detected in the polyclinic nurses at the highest rate of 72.22%.

As seen in Table 5, when the back region was taken as the dependent variable and the workstations were taken as the independent variable and desk job was chosen as the reference, according to the logistic regression analysis, the nurses in the service had a 1.90 times higher risk of back pain compared to the nurses working at a desk. In addition, nurses working in the outpatient clinics had a 4.31 times higher risk of back pain than nurses working at a desk.

In Table 6, the REBA scores of the jobs being done at eight workstations and the REBA scores of the workstations average scores are shown. The highest score in the work done is in the operating room nurses.

Table 7 shows the RULA scores of the jobs being done at eight workstations and the RULA average scores of the workstations.

Table 5. Logistic regression.

Workstation	OR (95% CI)		P
Service	1.90	(1.106–3.29)	.0020
Policlinic	4.31	(1.87–9.90)	<.0001

Table 8 shows the REBA scores and musculoskeletal pain rates of the works performed at eight workstations.

In addition, the REBA hospital average score and REBA hospital weighted average score are given.

As seen in Table 8, the risk scores of REBA scores are “high” at 10.00 in operating theaters and “moderate” (4–7) in all other stations. At the same time, the rate of musculoskeletal pain among nurses working in the operating room in the last year was 97.77%.

Table 9 shows the RULA scores and musculoskeletal pain rates of the works performed at eight workstations. In addition, the RULA hospital average score and RULA hospital weighted average score are given.

As seen in Table 9, the risk scores of RULA scores were “high” 7.00 in operating theaters and one level below this in all other stations. At the same time, the rate of musculoskeletal pain among nurses working in the operating room in the last year was 97.77%.

Discussion

Questionnaire forms should not be considered sufficient on their own due to the subjectivity of individuals. In a study, it was recommended to use a specific observation method in technical comparisons in measuring postural risks, and in turn to use it as a useful guide to precisely measure these postural stresses in the industry.²⁸ More accurate results can be obtained by applying different measurement-based techniques. In this direction, our aim in determining the risks of working postures in nurses working in the health sector is to reach more objective results and make more realistic arrangements by including REBA and RULA measurement methods, which are ergonomic risk assessment methods used in engineering fields, in addition to survey application.

In this study, BMI averages and distributions were analyzed. When the BMI distribution was examined in detail, it was seen that 3% of the nurses were underweight (BMI <18.50) and 11% were obese. The total distribution of these two groups is 14% and it can be considered that these two groups are at risk for musculoskeletal disorders and may have problems. In an article published in 2007, it was reported that patients with a BMI of 35 and above have many health problems in addition to pain complaints.¹ Another study showed that patients with high BMI also have a high rate of pain.²⁹ Again, the rate of health research in Turkey was found to be 21.1% among obese subjects in 2019.³⁰ When we look at the responses of the nurses in this study, we see that 92.46% of them have experienced musculoskeletal disorders in the last year. In an article published in 2020, it was stated that the most common musculoskeletal system complaints in working health professionals were in the lower back, back and leg regions.³¹ Another article published in 2018 stated that the most common body parts that felt

Table 6. REBA score values of jobs done at workstations.

Workstations	Service		Operating room		Emergency room		Polyclinic		Intensive care		Desk jobs		Lab		Invasive treatment	
	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA	REBA
The work done																
Desk job	5				5		5		5		5		5		5	
Nasogastric catheter					5				5							
Body temperature measurement	4				4		4		4							
Blood pressure measurement	5				5		5		5						5	
Vascular access	5				5				5				5		5	
Serum insertion	6				6				6						6	
Preparation of hemaf. solution									7							
Installing hemaf. solution									7							
Sterile nurse			10													
Circulating nurse			10													
Monitoring or EKG process	5				5		5		5				5			
Aspiration, Oxygen therapy	8				8				8							
Postural drainage	5				5				5							
Prepare the patient							9									
Aseptic applications	9				9				9				9		9	
Injection/Vaccination	5				5		5		5				5			
Positioning I	11				11				11							
Positioning II	11															
Positioning III	11															
REBA AVERAGE	6,92		10,00		6,08		5,50		6,21		5,00		5,80		6,00	

Table 7. RULA score values of jobs done at workstations.

Workstations	Service		Operating room		Emergency room		Policlinic		Intensive care		Desk jobs		Lab		Invasive treadmill	
	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA	RULA
The work done																
Desk job	5				5	5	5	5	5	5	5	5	5	5	5	5
Nasogastric catheter					4				4							
Body temperature measurement	3				3		3		3							
Blood pressure measurement	6				6		6		6						6	
Vascular access	4				4				4				4		4	
Serum insertion	5				5				5						5	
Preparation of hemafiltration solution																
Installing hemafiltration solution																
Sterile nurse (Operating room)		7														
Circulating nurse (Operating room)		7														
Monitoring or EKG process	6				6		6		6				6			
Aspiration, Oxygen support therapy	6				6				6							
Postural drainage	7				7				7							
Prepare the patient for examination																
Aseptic applications	7				7				7				7		7	
Injection/Vaccination administration	4				4				4				4			
Positioning I (Turning the patient)	7				7				7							
Positioning II (sitting position)	7															
Positioning III (raising the patient)	7															
REBA AVERAGE	5.69	7.00			5.33		5.16		5.57		5.00		5.20		5.40	

Table 8. Musculoskeletal complaints and REBA values in departments studied in the last 12 months.

Workstation	Nurse working at workstation (n)	In the last 12 months in musculoskeletal systems those who complain		Hospital workstations REBA average scores**	Reference REBA score	REBA degree	REBA precaution
		n	%				
Service	180	166	92.22	6.92	4-7	Middle	interventional treatment
Operating room	45	44	97.77	10.00	8-10	High	Needed in a short time
Emergency	16	16	100	6.08	4-7	Middle	Necessary
Polyclinic	24	18	75	5.50	4-7	Middle	Necessary
Intensive care	70	66	94.28	6.21	4-7	Middle	Necessary
Desk job	16	15	93.75	5.00	4-7	Middle	Necessary
Lab	20	19	95	5.80	4-7	Middle	Necessary
Interventional treatment	12	11	91.66	6.00	4-7	Middle	Necessary
Total	383	355	92.46	6.43**/6.86***	4-7	Middle	Necessary

*Row percentage.

**REBA Hospital average score.

***REBA Hospital weighted average score.

Table 9. Musculoskeletal complaints and RULA values at workstations worked in the last 12 months.

Workstation	Nurse working at workstation (n)	In the last 12 months in musculoskeletal systems those who complain		RULA avg. score	RULA precaution
		n	%*		
Service	180	166	92.22	5.69	The work should be reviewed and changes made quickly
Operating room	45	44	97.77	7.00	Change must be made urgently
Emergency	16	16	100	5.33	The work should be reviewed and changes made quickly
Polyclinic	24	18	75	5.16	The work should be reviewed and changes made quickly
Intensive care	70	66	94.28	5.57	The work should be reviewed and changes made quickly
Desk job	16	15	93.75	5.00	The work must be examined and changes made quickly
Lab	20	19	95	5.20	The work should be reviewed and changes made quickly
Interventional treatment	12	11	91.66	5.40	The work should be reviewed and changes made quickly
Total	383	355	92.46	5.54**/5.71***	The work should be reviewed and changes made quickly

*Row percentage.

**RULA Hospital average score.

***RULA Hospital weighted average score.

pain in intensive care nurses were the lower back in 88.3% and the back in 77.5%.³² In this study, 87.50% of the nurses working in the emergency department had lower back pain and 93.75% had back pain. In the intensive care workstation, 90.91% had lower back pain and 81.22% had back pain. Nurses on the service have a higher risk of experiencing back pain than desk workers. Additionally, nurses working in outpatient clinics were 4.31 times more likely to experience back pain than those working at a desk. It can be thought that this situation can be explained by the fact that the tasks performed at the stations, such as supporting patients' daily life activities and self-care practices, are more intense than other tasks.

In this study, the REBA hospital weighted mean score was 6.86 and the overall mean score was 6.43. When we compare the REBA average of these values according to the reference template, it is understood that it is in the range of (4–7). Again, it is seen that the risk level of this score is “medium” and that it is necessary to make changes as a precaution. This risk score in the REBA method and the level of precautions to be taken suggest that the high rate of nurses (92.46%) who stated that they had musculoskeletal disorders in the last year supports each other.

In terms of the department/station, the three highest REBA scores were found in the operating room (10.0), ward (6.92) and intensive care unit (6.21). As a result of REBA measurement in a group of operating room nurses in Tabriz in 2016, a risk score of 7.7 was found.³³ This risk level is close to the operating room risk level (10:high) in the study.

The nurses were asked about their musculoskeletal complaints such as pain, tingling and numbness. In the last year according to the departments they worked in, and the rates of nurses who said they had complaints according to the department they worked in were compared with the REBA scores in the departments they worked in. Accordingly, 97.77% of the operating room staff reported that they had such complaints in at least one region of the musculoskeletal system in the last year. Again, the REBA analysis score of the ward department, where 92.22% of the nurses working in the ward had musculoskeletal complaints in the last year, was 6.92 and the REBA score was in the range of 4–7; that is, it is clear that action is required in accordance with the definition of the precautionary situation as having a “moderate” risk level and that both the rate of responses to the questionnaire and the scores of the REBA method support each other. The rate of nurses working in intensive care, one of the departments with the third highest REBA score, experiencing the same complaints in the same time period is 94.28%. The result (6.21) determined when evaluated with the REBA technique for intensive care is also supportive of each other. For the rest of the working departments, the rate of complaints in the last twelve months does not fall below 75%. The REBA scores are also in the range of 4–7, which is considered moderate and it is necessary to make arrangements. In a public ambulance in Malaysia, they identified seven activities performed by employees and determined the risk levels for each of them with the REBA method. In the study, three activities were identified before the ambulance and their REBA scores were 4–7 “medium” risk and 8 “high” risk. Four activities were identified inside the ambulance and their scores were 4–6 “medium” risk and 10 “high” risk. Ambulance workers concluded that three of the seven defined activities were high risk.³⁴ In another study, the body risks of nurses working in five different intensive care units of a state hospital

were determined by the REBA method and scores in the range of 5–13 were obtained. It was reported that 63.3% of these risks were high risk. In this study, it was emphasized that 11 procedures were performed in the intensive care unit and 6 procedures had high risk values in REBA score.¹² In the study, 14 procedures were identified in the intensive care unit and 3 of them were found to have high risk scores. Again, in this study, moderate risk was determined in all but one location. In another similar study, 52.9% had moderate, 17.6% had high and 5.8% had very high risk scores in body posture evaluations of laboratory workers (specialist physicians, nurses, technicians, data processing personnel and cleaning personnel) with REBA method.³⁵ It can be considered that the studies have similar results because even if they are two different hospitals, it is natural that the results will be similar because the working sector is in the field of health and the work performed is similar. The effects of these similar risks on individuals may be different according to working hours, shifts and demographic characteristics.

If we consider the RULA method, it is actually the starting point of the REBA method. This method is designed to evaluate workers who are exposed to musculoskeletal stresses that cause upper limb (neck-shoulder-upper arm-lower arm-elbow-hand-wrist) complaints.³⁶ Although it works with similar principles to the REBA method in terms of logic, there are differences. The scoring range is between 1 and 7 and there are four impact-consequence levels.

In this study, the RULA hospital weighted average score was 5.71 and the overall average score was 5.54. When we compare these scores with the RULA average according to the reference template, it is understood that it is close to the 5–6 range. Thus, according to the impact level, it is concluded that “the work should be reviewed soon and changes should be made in the short term.” In a study examining the effects of hospital equipment on the posture of healthcare workers, researchers found that the RULA scores of physicians, surgeons and laboratory workers performing extremity examinations were 6.³⁷ These scores are similar to the scores in the study.

This risk score in the RULA method and the level of precautions to be taken suggest that the high rate of nurses (92.46%) who stated that they had musculoskeletal disorders in the last year support each other.

When analyzed in terms of the department/station, the three highest RULA scores were found to be 7 in the operating room, 5.69 in the ward and 5.57 in the intensive care unit. In our study, it was determined that the average RULA scores in other departments were all higher than 5. When the RULA scores of the other locations in the sections under review are assessed in the RULA reference template, the risk score is in the range 5–6, that is, impact level 3 should be applied (the work should be reviewed immediately and changes made in the short term). There is a high risk level according to the operating room score (7), and it is understood that the equivalent of this score as a measure according to the RULA reference template requires action. In this case, “It is necessary to urgently review the work and make changes.” The nurses were asked about their musculoskeletal complaints such as pain, tingling and numbness. In the last year according to the departments they worked in, and the rates of nurses who said they had complaints according to the department they worked in were compared with the RULA scores in the departments they worked in. Accordingly, 97.77% of the operating room nurses reported that they had such complaints in at least one region of the musculoskeletal system in the last year. Again, when we look at the RULA score (7) of the nurses

working in the operating room (RULA score range of 7 and above), it is understood that it is at a high level and there is a need to implement the definition of “urgently, the work done should be reviewed and changes should be made,” which corresponds to this score, which includes corrective actions to reduce risks in the ward. Again, the musculoskeletal complaints of the nurses working in the ward in the last year were 92.22% the RULA analysis score of the ward was 5.69 and the RULA score was in the 5–6 score range, which means that the precautionary situation with a moderate risk level should be carried out in accordance with the definition of “the work should be reviewed soon and changes should be made in the short term” and it is clear that both the rate of responses to the questionnaire and the scores of the RULA method support each other. In 2010, a RULA risk assessment was performed on a section of nurses working in the ward at a state university and they reached a score of 7, which is one level higher than the score of 5.69 found in our ward.²⁸ The scores in our study and in this study show that both risks require some ergonomic changes to be made, but measures to improve the body risks of nurses, which are the subject of this review in the literature, should be done “urgently.” In other words, unlike in this study, the change needs to be realized in a shorter period of time. The rate of nurses working in intensive care, one of the departments with the third highest RULA score in our study, experiencing the same complaints in the same time period is 94.28%. When evaluated with two different techniques for intensive care, it is seen that both results support each other again.

RULA scores were also found to be in the range of 5–6 and it is seen that it is necessary to make a regulation as “the work should be reviewed soon and changes should be made in the short term.” In a risk assessment analysis conducted on the working postures of dentists in Konya in 2016, RULA scores were found to be 6, that is, a moderate risk at rates similar to our study, and it was stated that the work should be reviewed ergonomically and changes should be made.³⁸

In the survey application, it is understood that the rates of nurses who stated that they had musculoskeletal disorders in the last year and the results of the analysis using the RULA method confirm each other. However, the average scores of the body risk analysis using the RULA method fall within the definition of a medium-level evaluation in all departments except the operating room, and it is seen that this level will definitely cause musculoskeletal disorders and that it is necessary to make arrangements, but when making general evaluations and comments, the existence of an upper level, which falls within the definition of the fourth impact level risk in the RULA reference table in the score range of 7 and above, and which is in the precautionary level of “urgently, the work should be reviewed and changes should be made” should not be ignored.

In a study published in 2022 in which nurses’ posture risk was calculated using the RULA method, 10% of subjects were at low risk, while 44% and 46% were at high and medium risk, respectively, of developing musculoskeletal disorders.³⁹

In a 2023 posture risk study of surgeons, the average RULA was 6.09 (4–7).⁴⁰

This similarity of evaluation is natural because REBA is derived from the RULA method. The REBA and RULA methods, which showed similar evaluations in our study, also show high risks in some studies and do not give completely healthy results.⁴¹

Fifty-nine percent of ergonomics experts recommend RULA and REBA analysis methods rather than other analysis methods.⁴²

Limitations of the study

In addition, since the research was conducted in a hospital with very intensive patient services, no pictures could be taken from all nurses. REBA and RULA score calculations were made by taking pictures from a total of 24 nurses, 3 from each workstation. In this case, we could not calculate REBA and RULA scores from every nurse we surveyed, and we could not make logistic regression comparisons. This was the most important limitation of the study.

Another disadvantage was the time-consuming nature of data collection by observation and questionnaires, and the fact that health care delivery is a time-competitive task. It is well known that nurses carry out their critical tasks in an environment where they have to be carried out rapidly and without hesitation as a whole.

Conclusions

As a result, the literature on this subject was examined and it was understood that the musculoskeletal complaints of the nurses were seen at levels that we can characterize as high, that is, above the medium values in both the survey results and the REBA and RULA score results.

It is seen that this study and the literature reviewed have similar results. The subjectivity of the data in surveys can affect the reliability of the results. Therefore, it is recommended that posture risks be measured at regular intervals not only by survey method but also by systematic observation methods in order to be cheap and objective. Although NIOSH, OWAS, etc. measurement methods are among the systematic observation methods, REBA and RULA methods have recently been preferred in the field of health due to their suitability.

In addition, after measurement determinations, ergonomic improvements and training should be organized for nurses, and working and shift hours should be updated. The necessary number of rest breaks must be arranged and a sufficient number of nurses must be employed.

The study is considered to be useful and further studies on this subject are recommended.

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